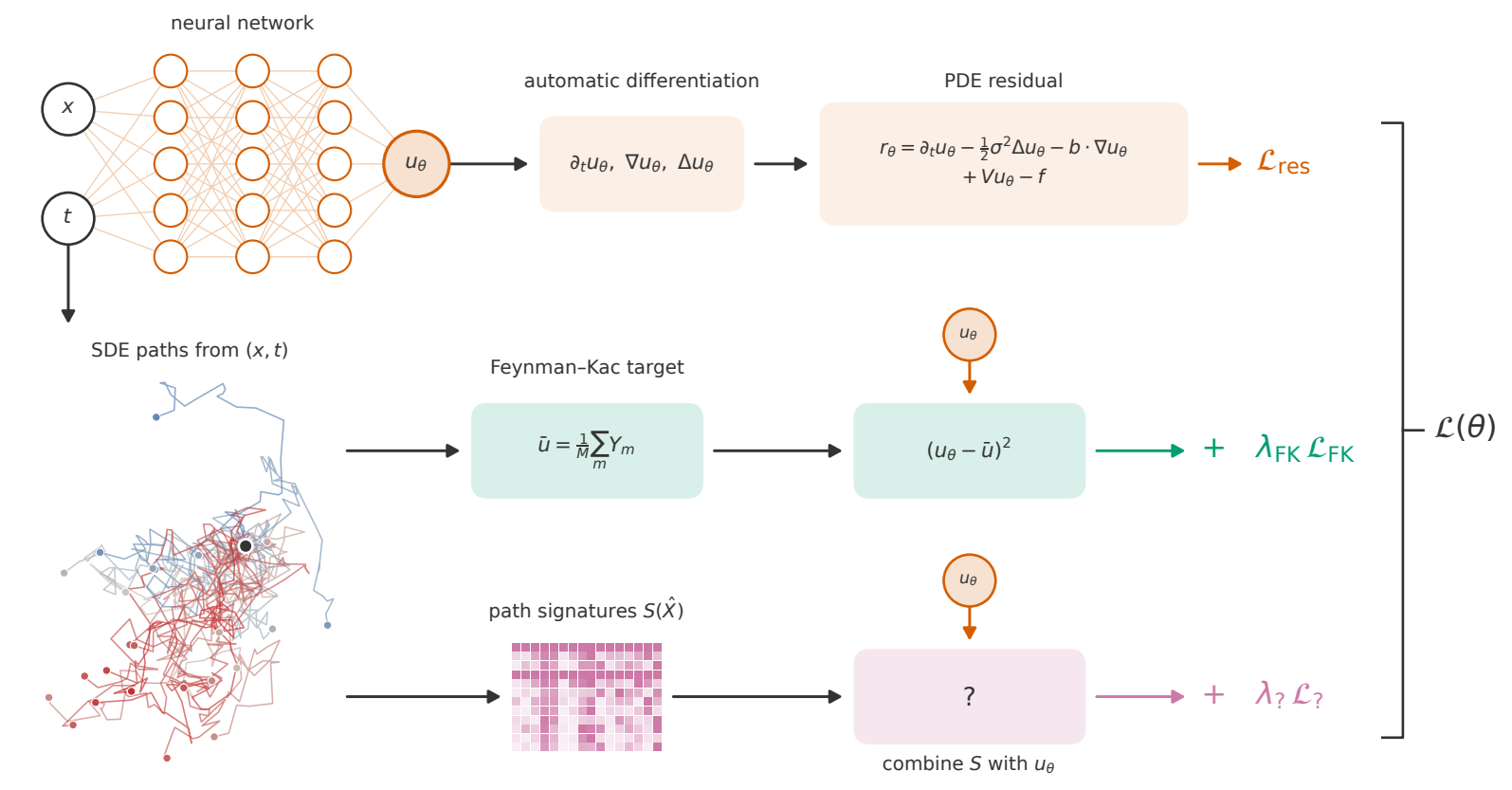
neural network
$x$
$t$
$u_\theta$
automatic differentiation
$\partial_t u_\theta,\ \nabla u_\theta,\ \Delta u_\theta$
PDE residual
$r_\theta = \partial_t u_\theta - \frac{1}{2}\sigma^2 \Delta u_\theta - b \cdot \nabla u_\theta + V u_\theta - f$
$\mathcal{L}_{\text{res}}$
SDE paths from $(x, t)$
Feynman–Kac target
$\bar{u} = \frac{1}{M}\sum_m Y_m$
$u_\theta$
$(u_\theta - \bar{u})^2$
$+\ \lambda_{\text{FK}}\,\mathcal{L}_{\text{FK}}$
path signatures $S(\hat{X})$
$u_\theta$
?
$+\ \lambda_?\,\mathcal{L}_?$
combine $S$ with $u_\theta$
$\mathcal{L}(\theta)$